

Package ‘GTAPViz’

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Title Visualization of GTAP Model Results from HAR and SL4 Files

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Description GTAPViz is an extension package of HARplus, designed to provide seamless data visualization for GTAP model results from ``.HAR`` and ``.SL4`` files. This package focuses on user-friendly visualization tools, enabling GTAP users to analyze and compare simulation results efficiently without requiring deep coding knowledge. With built-in functions for pivoting, mapping, unit assignments, and graphical representation, HARplusViz enhances economic model interpretation by generating structured, publication-ready plots. It is ideal for GTAP researchers, policymakers, and analysts looking for intuitive data exploration and visual insights from GEMPACK outputs.

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Encoding UTF-8

Roxygen list(markdown = TRUE)

RoxygenNote 7.3.2

BugReports <https://github.com/bodysbobb/GTAPViz/issues>

URL <https://bodysbobb.github.io/GTAPViz/>

Imports HARplus,

stats,
ggplot2,
dplyr,
tidyr,
openxlsx,
colorspace,
grDevices,
rlang,
scales,
purrr,
readxl,
writexl,
utils,
methods

Remotes github::Bodysbobb/HARplus

Suggests knitr,
rmarkdown,
devtools

VignetteBuilder knitr

Exports add_unit_col,
comparison_plot,
detail_plot,
macro_plot,
gtap_macros_data,
add_mapping_info

ImportFrom HARplus get_data_by_var load_sl4x
stats setNames
ggplot2 ggplot aes aes_string geom_bar geom_text geom_vline geom_col geom_hline facet_wrap
theme_minimal theme element_text element_rect element_blank unit margin labs
scale_fill_manual scale_y_continuous position_dodge expansion
dplyr filter mutate case_when
colorspace hex2RGB hex polarLUV sRGB
grDevices col2rgb rgb
rlang sym
scales squish

Depends R (>= 3.5)

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add_mapping_info	<i>Add Mapping Information to GTAP Data</i>
------------------	---

Description

Adds description and unit information to GTAP data based on specified mapping.

Usage

```
add_mapping_info(  
  data_list,  
  external_map = NULL,  
  mapping = "GTAPv7",
```

```

    description_info = TRUE,
    unit_info = TRUE
  )

```

Arguments

data_list	Data structure containing GTAP variables
external_map	Optional data frame with mapping information
mapping	Mapping mode: "GTAPv7" (default), "No", "Yes", or "Mix"
description_info	Logical. If TRUE, adds description information
unit_info	Logical. If TRUE, adds unit information

Value

Data structure with added mapping information, preserving original structure

Author(s)

Pattawee Puangchit

Examples

```

## Not run:
# Load example data
har_data <- HARplus::load_harx(system.file("extdata/in", "EXP1.sl4",
                                           package = "GTAPviz"))

# Corrected example with closing quotation
har_data <- add_mapping_info(har_data, mapping = "GTAPv7")

## End(Not run)

```

comparison_plot

Generate Comparison Plots

Description

Creates bar plots comparing values across categories with flexible panel configuration.

Usage

```

comparison_plot(
  data,
  plot_var = NULL,
  x_axis_from,
  split_by,
  title_prefix = "",
  title_suffix = "",
  compare_by_experiment = FALSE,

```

```

    separate_figure = FALSE,
    invert_panel = FALSE,
    description_as_title = FALSE,
    output_dir = NULL,
    panel_rows = NULL,
    panel_cols = NULL,
    color_tone = NULL,
    legend_position = "none",
    width = NULL,
    height = NULL
  )

```

Arguments

data	Data frame containing "Variable", "Value", "Experiment", "Unit" columns plus x_axis_from and split_by variables.
plot_var	Optional. Character vector or data frame with "Variable" column to specify which variables to plot.
x_axis_from	Column name to use for x-axis values (or panels if compare_by_experiment = TRUE).
split_by	Column name for separating plots (or panels if invert_panel = TRUE).
title_prefix	Optional text to prepend to plot titles.
title_suffix	Optional text to append to plot titles.
compare_by_experiment	If TRUE, Experiment is plotted on x-axis, x_axis_from values as panels.
separate_figure	If TRUE, creates individual figures for each panel value.
invert_panel	If TRUE, uses split_by for panels and Experiment for separation.
description_as_title	If TRUE, uses "Description" column for titles instead of "Variable".
output_dir	Optional path to save plots. If NULL, plots are only returned.
panel_rows	Optional number of panel rows. Auto-calculated if NULL.
panel_cols	Optional number of panel columns. Auto-calculated if NULL.
color_tone	Optional base color for the plot. NULL uses default ggplot2 colors.
legend_position	Position of legend: "none" (default), "bottom", "top", "left", "right".
width	Optional width in inches. Auto-calculated if NULL.
height	Optional height in inches. Auto-calculated if NULL.

Value

A list of ggplot objects or a single ggplot object depending on settings

Author(s)

Pattawee Puangchit

Examples

```
## Not run:
# Generate comparison plot
comparison_plot(
  data = sl4.plot.data$REG,
  x_axis_from = "Region",
  plot_var = sl4plot,
  title_prefix = "Impact on",
  output_dir = output.folder,
  compare_by_experiment = FALSE,
  description_as_title = TRUE,
  separate_figure = FALSE,
  color_tone = "grey",
  width = 20,
  height = 12,
  legend_position = "bottom"
)

## End(Not run)
```

convert_units

*Convert Multiple Units in Nested Data Structures***Description**

Convert Multiple Units in Nested Data Structures

Usage

```
convert_units(
  data,
  change_unit_from,
  change_unit_to,
  adjustment,
  value_col = "Value",
  unit_col = "Unit"
)
```

Arguments

data	A data structure (list, data.frame, or nested combination)
change_unit_from	Character vector of units to change (case-insensitive)
change_unit_to	Character vector of new units (same length as change_unit_from)
adjustment	Character vector of operations or numeric vector (same length as change_unit_from)
value_col	Column name containing values to adjust (default: "Value")
unit_col	Column name containing unit information (default: "Unit")

Value

Data structure with same format as input but with adjusted values and units

Author(s)

Pattawee Puangchit

Examples

```
## Not run:
# Load example data
har_data <- HARplus::load_harx(system.file("extdata/in", "EXP1.sl4",
                                           package = "GTAPViz"))

har_data <- convert_units(har_data,
  change_unit_from = c("million USD"),
  change_unit_to = c("billion USD"),
  adjustment = c("/1000"))

## End(Not run)
```

decomp_table	Create Decomposition Table with Multi-Data Support
--------------	--

Description

Processes decomposition data, allowing multiple data frames as input. Converts specified columns into wide format per dataset while supporting multiple headers, independent column selections, and optional total calculations.

Usage

```
decomp_table(
  data_list,
  header,
  wide_cols,
  total_column = FALSE,
  export_table = FALSE,
  multi_sheet = FALSE,
  output_dir = NULL,
  sheet_names = NULL,
  include_units = FALSE
)
```

Arguments

data_list	A named list of data frames. Each data frame must contain an "Experiment" column.
header	A character vector specifying which dataset names from data_list should be processed.
wide_cols	A named list specifying the column to be transformed into wide format for each dataset. Example: list(A = "COLUMN", E1 = "PRICES").
total_column	Logical. If TRUE, calculates a total column for each dataset by summing numeric columns. Default is FALSE.

export_table	Logical. If TRUE, exports the result to an Excel file. Default is FALSE.
multi_sheet	Logical. If TRUE, each dataset is written to a separate sheet in the Excel file. Default is FALSE.
output_dir	Optional character string specifying the directory where the Excel file should be saved.
sheet_names	Optional named list providing custom names for sheets in the exported Excel file.
include_units	Logical. If TRUE and a "Unit" column exists in the data, unit information is merged into the final output. Default is FALSE.

Details

- If data is a single data frame, header is ignored, and wide_cols should be a single column name.
- If data is a list of data frames, header must be specified to select relevant subsets of each dataset.
- wide_cols must be a named list matching the names of data (e.g., "A", "E1").
- When total_column = TRUE, numerical values in each row are summed into a "Total" column.
- If export_table = TRUE, the function saves the result to an Excel file, either as a single sheet or multiple sheets (if multi_sheet = TRUE).

Value

A list of data frames, each containing wide-format decomposition data. If export_table = TRUE, the results are also saved to an Excel file.

Author(s)

Pattawee Puangchit

Examples

```
## Not run:
# Example list of data frames
har.plot.data <- list(
  A = data.frame(Experiment = c("Base", "Policy"),
    Header = "A", COLUMN = c("Tariff", "NTM"), Value = c(10, 5)),
  E1 = data.frame(Experiment = c("Base", "Policy"),
    Header = "E1", PRICES = c("GDP", "Consumption"), Value = c(105, 55))
)

# Convert decomposition tables for both datasets
result <- decomp_table(data_list = har.plot.data, header = c("A", "E1"),
  wide_cols = list(A = "COLUMN", E1 = "PRICES"), total_column = TRUE)

# Export as an Excel file with multiple sheets
decomp_table(data_list = har.plot.data, header = c("A", "E1"),
  wide_cols = list(A = "COLUMN", E1 = "PRICES"),
  total_column = TRUE, export_table = TRUE, multi_sheet = TRUE,
  output_dir = "results", sheet_names = list(A = "Welfare", E1 = "Prices"))

## End(Not run)
```

detail_plot

Generate Detailed Plot

Description

Creates a detailed plot reporting all regions or sectors depending on the variable selection. Uses a separate dataframe to specify variables to plot and their titles.

Usage

```
detail_plot(
  data,
  plot_var = NULL,
  y_axis_from,
  split_by,
  title_prefix = "",
  title_suffix = "",
  description_as_title = TRUE,
  panel_rows = NULL,
  panel_cols = NULL,
  positive_color = "#2E8B57",
  negative_color = "#CD5C5C",
  output_dir = NULL,
  top_impact = NULL,
  width = NULL,
  height = NULL,
  separate_figure = FALSE,
  invert_panel = FALSE,
  legend_position = "none",
  width_dodge = 0.5,
  width_bar = 0.4
)
```

Arguments

data	A data frame containing the full dataset.
plot_var	A data frame with "Variable" and "PlotTitle" columns specifying which variables to plot and their titles. If NULL, all variables in the data will be plotted with default titles.
y_axis_from	Character vector. Column names to use for the y-axis.
split_by	Character. Column used to separate different plots.
title_prefix	Optional character string to prepend to the title.
title_suffix	Optional character string to append to the title.
description_as_title	Logical. If TRUE, uses the "Description" column from plot_var as the plot title.
panel_rows	Optional numeric. Number of panel rows. If left blank, it will be automatically adjusted.
panel_cols	Optional numeric. Number of panel columns. If left blank, it will be automatically adjusted.

positive_color	Character. Color for positive values.
negative_color	Character. Color for negative values.
output_dir	Optional character. Directory to save the output plot.
top_impact	Optional numeric. Number of top impact factors to include. If left blank, all factors will be included.
width	Optional numeric. Width of the output plot. If left blank, it will be automatically adjusted.
height	Optional numeric. Height of the output plot. If left blank, it will be automatically adjusted.
separate_figure	Logical. If TRUE, creates separate figures for each combination of variable and category.
invert_panel	Logical. If TRUE, swaps the roles of Experiment and split_by in faceting and figure separation.
legend_position	Character. Position of the legend: "none" (default), "bottom", "top", "left", or "right".
width_dodge	Numeric. Dodge width for bar position adjustment. Default is 0.5.
width_bar	Numeric. Width of bars in the plot. Default is 0.4.

Value

A ggplot object or NULL if no plots could be generated.

Author(s)

Pattawee Puangchit

gtap_macros_data	<i>Extract and Aggregate Scalar Macroeconomic Variables</i>
------------------	---

Description

Extracts scalar macroeconomic variables from multiple SL4 datasets and aggregates them into a structured data frame.

Usage

```
gtap_macros_data(
  input_dir = NULL,
  experiment_names = NULL,
  output_dir = NULL,
  output_formats = NULL,
  subtotal_level = FALSE,
  select_var = NULL
)
```

Arguments

Character. Directory containing SL4 files.
 experiment_names Character vector. List of experiment names corresponding to SL4 files.
 output_dir Character (optional). Directory to save exported data.
 output_formats Character vector (optional). List of output formats (e.g., "csv", "xlsx").
 subtotal_level Logical. Whether to include subtotal levels in the processed data.
 select_var Character vector (optional). List of specific variable names to filter from the final result. If NULL, all variables are returned.

Value

A sorted dataframe containing processed GTAP macro data.

Author(s)

Pattawee Puangchit

See Also

[add_mapping_info](#), [process_gtap_data](#)

Examples

```
## Not run:
# Method 1: Extract all variables
input_dir <- paste0(input.folder)
experiment <- c("EXP1", "EXP2") # File name (.sl4)
Macros <- gtap_macros_data(input_dir, experiment_names = experiment,
  subtotal_level = FALSE)

# Method 2: Filter specific variables
Macros <- gtap_macros_data(input_dir, experiment_names = experiment,
  select_var = c("qgdp", "pop", "gdpxp"))

## End(Not run)
```

macro_plot

Generate Macro Plot with Flexible Title Options

Description

Creates figures of macro variables from the Macros header in the SL4. Uses a separate dataframe to specify variables to plot with options for using either variable names, PlotTitle, or Description columns for plot titles.

Usage

```
macro_plot(
  data,
  plot_var = NULL,
  title_prefix = "",
  title_suffix = "",
  compare_by_experiment = FALSE,
  description_as_title = FALSE,
  output_dir = NULL,
  panel_rows = NULL,
  panel_cols = NULL,
  color_tone = NULL,
  separate_figure = FALSE,
  width = NULL,
  height = NULL,
  legend_position = "none"
)
```

Arguments

<code>data</code>	A data frame containing the full dataset with macroeconomic variables.
<code>plot_var</code>	A data frame with "Variable" column and optionally "PlotTitle" or "Description" columns specifying which variables to plot and their titles. If NULL, all variables in the data will be plotted with default titles.
<code>title_prefix</code>	Optional character string to prepend to the title.
<code>title_suffix</code>	Optional character string to append to the title.
<code>compare_by_experiment</code>	Logical. If TRUE, compares experiments within x-axis categories.
<code>description_as_title</code>	Logical. If TRUE, uses the "Description" column for plot titles instead of "PlotTitle". Default is FALSE.
<code>output_dir</code>	Optional character. Directory to save the output plot.
<code>panel_rows</code>	Optional numeric. Number of panel rows. If left blank, it will be automatically adjusted.
<code>panel_cols</code>	Optional numeric. Number of panel columns. If left blank, it will be automatically adjusted.
<code>color_tone</code>	Optional character. Base color for the plot. If left blank, a default color will be used.
<code>separate_figure</code>	Logical. If FALSE, combines multiple variables into one plot.
<code>width</code>	Optional numeric. Width of the output plot. If left blank, it will be automatically adjusted.
<code>height</code>	Optional numeric. Height of the output plot. If left blank, it will be automatically adjusted.
<code>legend_position</code>	Position of the legend: "none", "bottom", "top", "left", or "right"

Value

A ggplot object or a list of plots if `separate_figure = TRUE`.

Author(s)

Pattawee Puangchit

Examples

```
## Not run:
macro_plot(Macros,
            plot_var = macro.map,
            compare_by_experiment = FALSE,
            color_tone = "grey",
            output_dir = output.folder,
            description_as_title = FALSE,
            panel_rows = 2,
            width = 45,
            separate_figure = FALSE,
            legend_position = "bottom")

## End(Not run)
```

process_gtap_data

*Process GTAP Data Automation with Flexible Output Options***Description**

Processes GTAP data from SL4 and HAR files with options for exporting and preparing plot-ready data.

Usage

```
process_gtap_data(
  experiment,
  mapping_info = "GTAPv7",
  project_dir = NULL,
  input_dir = NULL,
  output_dir = NULL,
  output_formats = NULL,
  plot_data = FALSE,
  region_select = NULL,
  sector_select = NULL,
  subtotal = FALSE,
  sl4_list_name = "sl4.plot.data",
  har_list_name = "har.plot.data",
  sl4_structure_name = "sl4.structure",
  har_structure_name = "har.structure",
  GTAPMacro_name = "GTAPMacro",
  sl4var = NULL,
  harvar = NULL,
  sl4map = NULL,
  harmap = NULL
)
```

Arguments

experiment	Character vector. Case names to process.
mapping_info	Character. Mapping mode: "GTAPv7" (default), "Yes", "No", or "Mix".
project_dir	Character. Path to the project folder with "in" and "out" subfolders.
input_dir	Character. Path to the input folder. Overrides project_dir/in if specified.
output_dir	Character. Path to the output folder. Overrides project_dir/out if specified.
output_formats	Character vector or list. Exports data in these formats (valid: "csv", "stata", "rds", "txt").
plot_data	Logical. If TRUE, prepares data for plotting and assigns to variables.
region_select	Optional vector. Specifies regions to filter the data.
sector_select	Optional vector. Specifies sectors to filter the data.
subtotal	Logical. If TRUE, includes subtotal data. Default is FALSE.
sl4_list_name	Character. Variable name for SL4 plotting data if generating plot data. Default is "sl4.plot.data".
har_list_name	Character. Variable name for HAR plotting data if generating plot data. Default is "har.plot.data".
sl4_structure_name	Character. Variable name for SL4 structure if generating plot data. Default is "sl4.structure".
har_structure_name	Character. Variable name for HAR structure if generating plot data. Default is "har.structure".
GTAPMacro_name	Character. Variable name for GTAP macro data if generating plot data. Default is "GTAPMacro".
sl4var	Data frame, NULL, or FALSE. Variables to extract from SL4 files. Set to NULL to extract all variables, or FALSE to skip SL4 processing.
harvar	Data frame, NULL, or FALSE. Variables to extract from HAR files. Set to NULL to extract all variables, or FALSE to skip HAR processing.
sl4map	Data frame or NULL. Mapping information for SL4 variables (with "Variable", "Description", and "Unit" columns).
harmap	Data frame or NULL. Mapping information for HAR variables (with "Variable", "Description", and "Unit" columns).

Details

This function provides a complete automation workflow for processing GTAP model outputs, with flexible output options and optional filtering for plot data.

The key parameters `sl4var` and `harvar` each accept three different input types:

- **Data frame:** Contains variable mappings with required "Variable" column. When provided, only specified variables will be extracted.
- **NULL:** Extracts all available variables from the respective file type.
- **FALSE:** Completely skips processing of that file type, allowing the function to focus only on the other file type.

The `mapping_info` parameter controls how descriptions and units are assigned:

- **GTAPv7:** Uses standard GTAPv7 definitions (default)
- **Yes:** Uses only the supplied descriptions and units from sl4map/harmap
- **No:** Does not add any descriptions or units
- **Mix:** Prioritizes supplied descriptions and units, falling back to GTAPv7 for any missing values

Value

Returns the processed data invisibly, which will not be printed to the console.

Author(s)

Pattawee Puangchit

See Also

[add_mapping_info](#), [gtap_macros_data](#)

Examples

```
## Not run:
# Extract data with region and experiment filters
process_gtap_data(
  sl4var = NULL,
  harvar = NULL,
  sl4map = sl4map,
  harmap = harmap,
  region_select = selected_regions,
  sector_select = NULL,
  subtotal = FALSE,
  experiment = experiment,
  mapping_info = info.mode,
  project_dir = project.folder,
  plot_data = TRUE,
  output_formats = list(
    "csv" = "No",
    "stata" = "No",
    "rds" = "No",
    "txt" = "No"))

## End(Not run)
```

rename_GTAP_bilateral *Rename GTAP Bilateral Trade Columns*

Description

Renames bilateral trade columns in GTAP data output

Usage

```
rename_GTAP_bilateral(data)
```

Arguments

data Data structure containing GTAP bilateral trade data

Value

Data structure with renamed bilateral trade columns

Author(s)

Pattawee Puangchit

Examples

```
## Not run:
# Load Sample Data
sl4_data <- HARplus::load_sl4x(system.file("extdata/in", "EXP1-WEL.sl4",
                                           package = "GTAPViz"))

sl4_data <- rename_GTAP_bilateral(df)

## End(Not run)
```

rename_value	<i>Rename Values in Data Structures</i>
--------------	---

Description

Replaces specified values in a column across nested data structures

Usage

```
rename_value(data, column_name = NULL, mapping.file)
```

Arguments

data Data structure containing data to rename

column_name Column name to modify. If NULL, extracted from mapping file

mapping.file Data frame with OldName and NewName columns for renaming

Value

A modified data frame or list of data frames with specified values replaced.

Author(s)

Pattawee Puangchit

Examples

```
## Not run:
# Example mapping file
wefare.decomp.rename <- data.frame(
  OldName = c("alloc_A1", "ENDWB1", "tech_C1"),
  NewName = c("Allocation", "Endowment", "Technology"),
  ColumnName = "Variable",
  stringsAsFactors = FALSE
)

# Load example data
har_data <- HARplus::load_harx(system.file("extdata/in", "EXP1.sl4",
                                           package = "GTAPViz"))

# Apply renaming
modified_data <- rename_value(har_data, mapping.file = wefare.decomp.rename)

## End(Not run)
```

report_table

Create Report Table

Description

Generates a structured report table from a given dataset, allowing for flexible formatting and customization of column names. The function reshapes the data to present variables in a wide format with experimental groups as rows.

Usage

```
report_table(
  data,
  vars,
  x_axis_from,
  output_dir = NULL,
  workbook_name = NULL,
  sheet_name = "Results",
  col_names = NULL
)
```

Arguments

data	A data frame or list containing the dataset. If a list, it is converted into a data frame before processing.
vars	A character vector specifying the variables to include in the report.
x_axis_from	A character string specifying the column used for grouping (e.g., "REG", "COMM").
output_dir	Optional character string specifying the directory where the report will be saved as an Excel file. If NULL, the table is not exported.
workbook_name	Optional character string for the Excel workbook name. Default is "Report_table".
sheet_name	Optional character string for the worksheet name. Default is "Results".

col_names Optional data frame or character vector for renaming columns. If a data frame, it must contain "Variable" and "PlotTitle" columns to map variable names to custom titles. If a character vector, it must match the length of vars.

Details

- The function checks for the required columns "Experiment", "Variable", "Value", and the specified x_axis_from column.
- The table is formatted to ensure numerical values are rounded to two decimal places.
- If output_dir is provided, the table is exported as an Excel file with enhanced formatting.
- If column renaming (col_names) is specified, it is applied to improve readability.

Value

A formatted data frame in wide format with grouped variables as columns and experimental groups as rows. If output_dir is provided, the function also saves the output to an Excel file.

Author(s)

Pattawee Puangchit

See Also

[scalar_table](#), [decomp_table](#)

Examples

```
## Not run:
# Example dataset
data <- data.frame(
  Experiment = rep(c("Baseline", "Policy"), each = 3),
  REG = rep(c("USA", "CHN", "EU"), 2),
  Variable = rep(c("GDP", "Consumption", "Investment"), 2),
  Value = c(100, 50, 30, 105, 55, 35)
)

# Generate report table
report <- report_table(data, vars = c("GDP", "Consumption"),
  x_axis_from = "REG", output_dir = "results")

# Custom column names
col_names_df <- data.frame(
  Variable = c("GDP", "Consumption"),
  PlotTitle = c("Gross Domestic Product", "Household Consumption")
)

report_table(data, vars = c("GDP", "Consumption"), x_axis_from = "REG",
  output_dir = "results", col_names = col_names_df)

## End(Not run)
```

scalar_table	Create Scalar Result Table
--------------	----------------------------

Description

Generates a formatted table summarizing scalar results from a given dataset. The function transforms long-format data into a wide format with scenarios as columns, making comparisons across experiments more accessible.

Usage

```
scalar_table(
  data,
  vars = NULL,
  output_dir = NULL,
  workbook_name = "scalar_results",
  sheet_name = "Results"
)
```

Arguments

data	A data frame or list containing scalar data, where each observation consists of an experiment name, variable name, and corresponding value.
vars	Optional character vector of variable names to include in the table. If NULL, all available variables in the dataset will be used.
output_dir	Optional character. Directory where the generated table will be saved as an Excel file. If NULL, no file will be created.
workbook_name	Optional character. Name of the output Excel workbook. Default is "scalar_results".
sheet_name	Optional character. Name of the worksheet in the Excel file. Default is "Results".

Details

- If data is a list and not a data frame, the function attempts to combine it into a data frame.
- The table is formatted to ensure numerical values are rounded to two decimal places.
- If output_dir is specified, the table is exported as an Excel file with proper formatting.
- Required columns in data are "Experiment", "Variable", and "Value".

Value

A formatted data frame in wide format with variables as rows and experiments as columns. If output_dir is provided, the function also saves the output to an Excel file.

Author(s)

Pattawee Puangchit

See Also

[report_table](#), [decomp_table](#)

Examples

```
## Not run:
sl4_data1 <- HARplus::get_data_by_var(c("globalcgds", "vgdpwld", "WEV"), sl4_data)
scalar_table <- scalar_table(
  data = sl4_data1,
  vars = "vgdpwld",
  output_dir = temp_dir,
  workbook_name = "global_indicators",
  sheet_name = "Results"
)

## End(Not run)
```

stack_plot

Generate Stacked Bar Plot with Auto-Dimensions

Description

Creates stacked bar plots with automatic total calculations and flexible grouping. Uses a separate dataframe to specify variables to plot and their titles. Auto-adjusts dimensions and panel layout if not specified.

Usage

```
stack_plot(
  data,
  plot_var = NULL,
  x_axis_from,
  stack_value_from,
  title_prefix = "",
  title_suffix = "",
  y_axis_title = NULL,
  compare_by_experiment = FALSE,
  separate_figure = FALSE,
  unstack_plot = FALSE,
  output_dir = NULL,
  panel_rows = NULL,
  panel_cols = NULL,
  color_tone = NULL,
  show_total = TRUE,
  description_as_title = TRUE,
  legend_position = "bottom",
  y_limit = NULL,
  width = NULL,
  height = NULL
)
```

Arguments

data A data frame containing the full dataset.

plot_var	A data frame with "Variable" and "PlotTitle" columns specifying which variables to plot and their titles. If NULL, all variables in the data will be plotted with default titles.
x_axis_from	Character. Column name to use for x-axis grouping
stack_value_from	Character. Column name containing the stack categories
title_prefix	Optional character string to prepend to the title
title_suffix	Optional character string to append to the title
y_axis_title	Character. Y-axis label. If NULL, uses Unit if available.
compare_by_experiment	Logical. If TRUE, compares x-axis values within experiments
separate_figure	Logical. If TRUE, creates separate figures instead of panels
unstack_plot	Logical. If TRUE, displays components side by side instead of stacked and creates separate figures for each x_axis_from value
output_dir	Optional character. Directory to save the output plot
panel_rows	Optional numeric. Number of panel rows
panel_cols	Optional numeric. Number of panel columns
color_tone	Optional character. Base color for plot color generation
show_total	Logical. If TRUE, displays total values above stacked bars. Default is TRUE.
description_as_title	Logical. If TRUE, uses the "Description" column as the title instead of "Variable". Default is TRUE.
legend_position	Character. Position of the legend: "bottom" (default), "top", "left", "right", or "none"
y_limit	Optional numeric vector. Custom y-axis limits (e.g., c(-5, 5))
width	Optional numeric. Width of the output plot
height	Optional numeric. Height of the output plot

Value

A list of ggplot objects or a single ggplot object depending on settings

Author(s)

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